

- At Microsoft Build 2024, there was the announcement of Copilot for Azure Sql going to public preview. Now the question arises, what can this type of Copilot do for me as a DBA, Developer or any other role where I'm working with Azure Sql.
- To help you get started, this session will introduce Copilot for Sql to you. Let's get into SSMS, find this Copilot and let it work a bit for us. Let's dig into general questions, data related questions but also database management!
- Don't expect deep dives into the inner workings, but I will get a bit into the security as this is paramount in any data resource.
- Buck Woody once (often) said: "don't type in demos". Join this session to see me break that rule! And even better, send in your own questions and see what happens!

Copilot for SQL

The background of the slide is a dark purple field filled with a complex pattern of small, semi-transparent dots in shades of purple and green. Overlaid on this are thin, wavy lines in green and purple, and a series of overlapping, semi-transparent rectangular outlines in a light green color, creating a layered, abstract effect.

Reitse Eskens

What the
query is
this?!?!



Thank you to our sponsors.
Please say hello to them and find
out how they can help you.

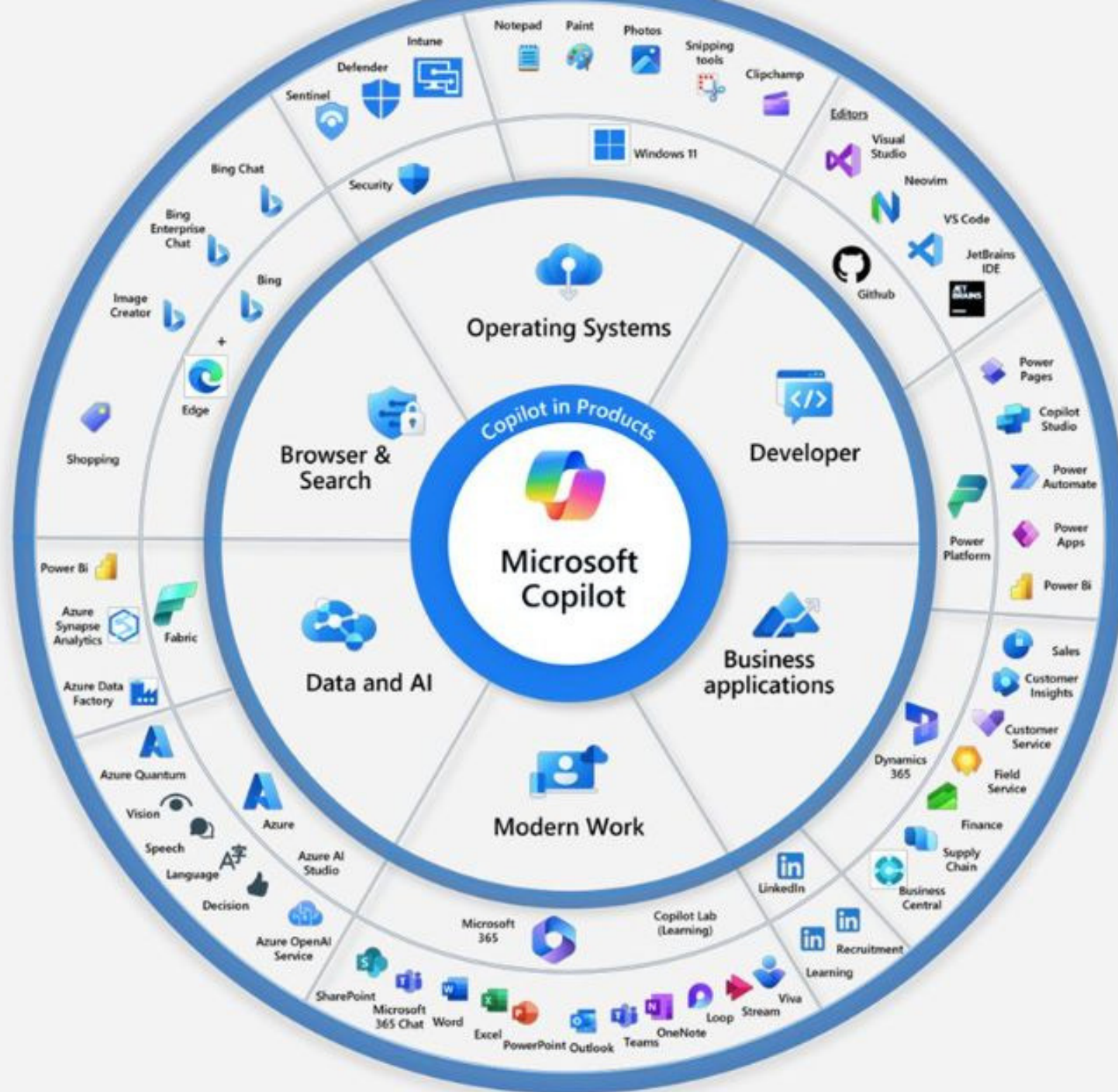




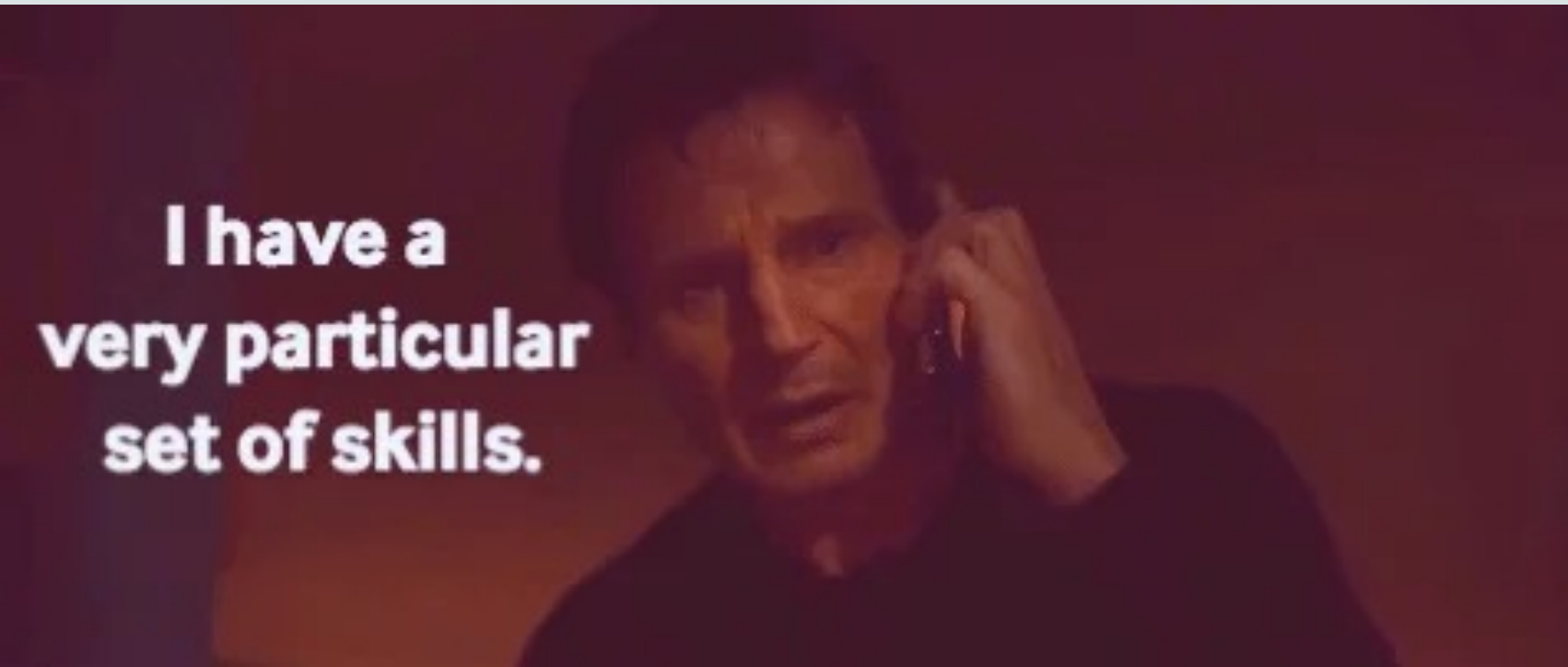
Another session on
Copilot?

A close-up photograph of a hand holding a camera lens. The lens is dark and has many concentric rings. The background is bright and out of focus, with some bokeh light effects. The text is overlaid on the lens.

Azure Copilot is GA
Copilot for Fabric SQL GA
Copilot in SSMS is GA (some new parts in
preview



How many
Copilots are
there?



ForEach
(Copilot in
Copilots)

COPILOT

NOT YOUR OTTO PILOT



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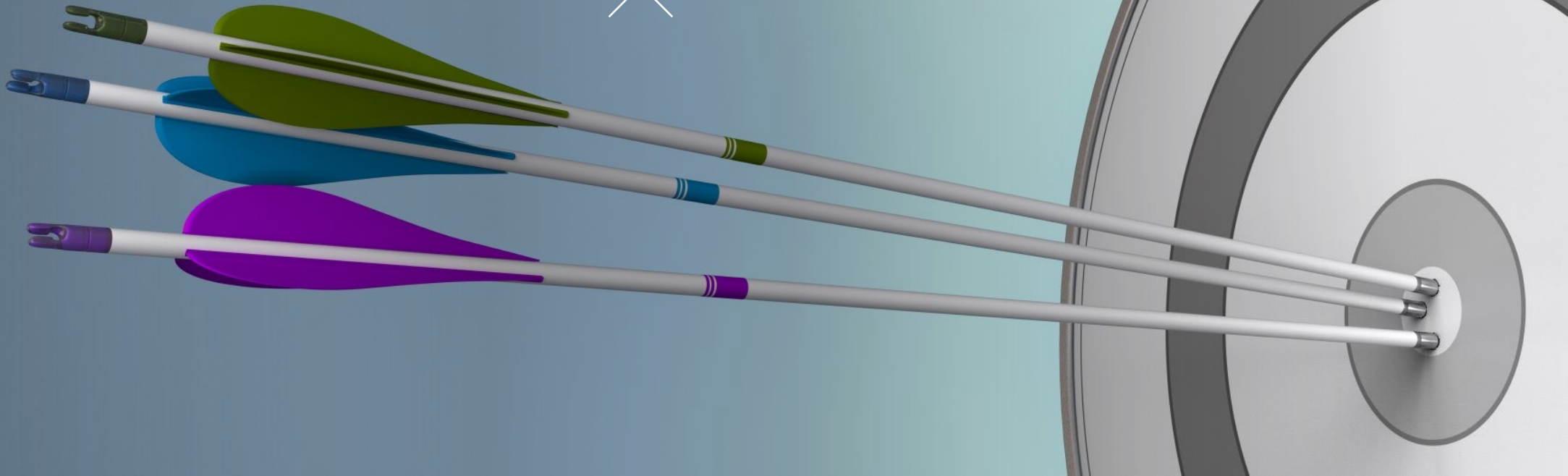
<https://sqlreitse.com>



@2meterDBA



Is Sql Copilot for you?





What about
security?



Differences between
Sql Copilots



Azure Sql Copilot

Available for Azure SQL

You can use it in the portal

No database context

The image features a dense field of three-dimensional, dark grey dollar signs (\$). In the center of the frame, a single, bright orange question mark (?) stands out prominently. The text "Portal questions" is written in a white, serif font, positioned horizontally across the middle of the image, partially overlapping the orange question mark and the surrounding dollar signs.

Portal questions



What about
security?

assignments - sql-planes-preview

Current role assignments

Eligible assignments

Assignments for the selected user, group, service principal, or managed identity at this scope or inherited to this scope.

🔍 Search by assignment name or description

Role assignments (1) ⓘ

Role	Description	Scope	Group assignment	Condition
Reader	View all resources, but does not ...	This resource	SG SQL Readers	None

Based on the error message you received, it appears that the login failed because the Entra ID principal being used is not authorized to access the 'TestDB' database. This could be due to a variety of reasons such as incorrect credentials, insufficient permissions, or the user not being added to the database.

To resolve this issue, you might need to:

1. **Verify the credentials:** Ensure that the username and password are correct. If you're unsure, you might need to reset the password or create a new login.

assignments - sql-planes-preview

Current role assignments

Eligible assignments

Assignments for the selected user, group, service principal, or managed identity at this scope or inherited to this scope.

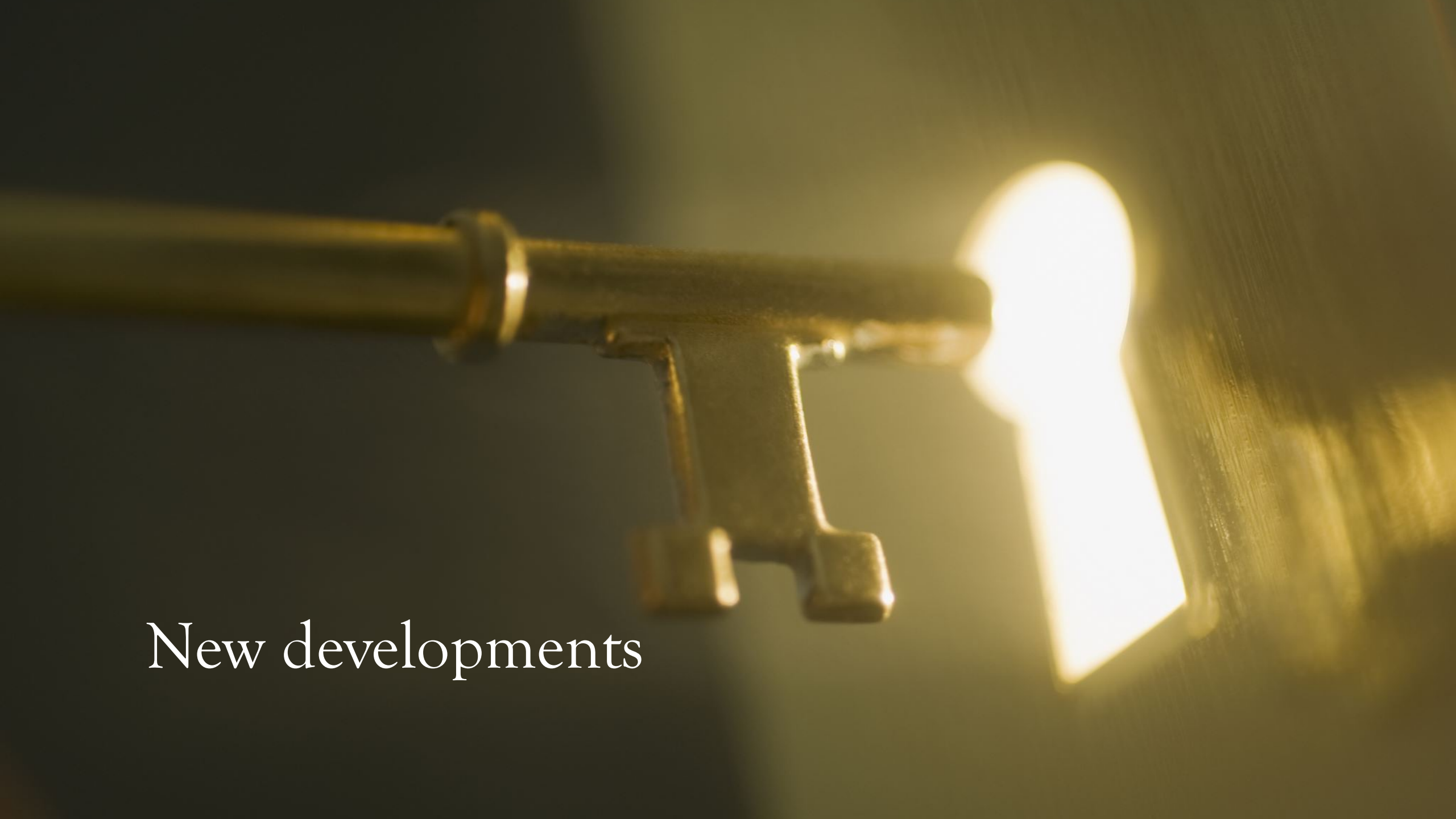
Search by assignment name or description

Role assignments (1)

Role	Description	Scope	Group assignment	Condition
Owner	Grants full access to manage all r...	This resource	SQ Sql Owners	View/Edit

I'm sorry, but I can't provide specific information about the database you're asking about because the login attempt for the user failed. The Entra ID principal being used is not authorized to access this database. This means the user credentials provided do not have the necessary permissions to access the database. However, I can provide some general information about Azure SQL Database, a fully managed platform as a service (PaaS) database engine. Here are some key features:





New developments

Guidelines, and a specific Copilot account



Copilot Instructions



What (not) to do



How to return
results

Github Copilot Execution context

Create an account
specifically for
Copilot



Make it known in
the database
context for
Copilot



It is now limited
to what the
account can do

Github
Copilot
Execution
context

[https://techcommunity.microsoft.com/
blog/sqlserver/introducing-execution-
context-for-github-copilot-in-
ssms/4527272](https://techcommunity.microsoft.com/blog/sqlserver/introducing-execution-context-for-github-copilot-in-ssms/4527272)

Let's do the live demo!



Recap

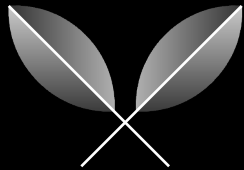




Check, please!

#Laff

Any Questions?



Use an extended event to see for
yourself!



What can you tell me about this database?

```
1  SELECT
2    @@SERVERNAME AS Server,
3    @@VERSION AS SQLServerVersion,
4    d.name AS DBName,
5    d.recovery_model_Desc AS RecoveryModel,
6    d.Compatibility_level AS CompatiblityLevel,
7    d.create_date,
8    d.state_desc,
9    slo.*
10 FROM sys.databases AS d
11 INNER JOIN sys.database_service_objectives AS slo
12 ON d.database_id = slo.database_id;
```

Opening the Azure Portal Query editor (preview)

```
<value><![CDATA[SELECT CONCAT(QUOTENAME(TABLE_SCHEMA), '.', QUOTENAME(TABLE_NAME))  
FROM INFORMATION_SCHEMA.TABLES  
WHERE TABLE_SCHEMA <> 'SYS']]></value>
```

Opening the Azure Portal Query editor (preview)

```
<value><![CDATA[SELECT  
    s.NAME AS SchemaName,  
    t.NAME AS TableName  
FROM sys.tables t  
INNER JOIN sys.schemas s  
ON t.schema_id = s.schema_id  
ORDER BY 1, 2]]></value>
```


Building the internal dataset

```
--
13 SELECT CONCAT(QUOTENAME(TABLE_SCHEMA), '.', QUOTENAME(TABLE_NAME)),
14             QUOTENAME(COLUMN_NAME)
15 FROM INFORMATION_SCHEMA.COLUMNS
16 WHERE TABLE_SCHEMA <> 'SYS'
17        AND CONCAT(QUOTENAME(TABLE_SCHEMA), '.', QUOTENAME(TABLE_NAME)) IN
18        ('[dbo].[_TransactionIndex_31adb587-a9ae-4f15-839b-1bb9ac305049]', '[dbo].[EmployeeHoliday]', '[dbo].[EmployeeSalary]',
19         '[dbo].[pivottable]', '[STA].[AIRPORTS]', '[STA].[CARRIERS]', '[STA].[FLIGHTDATA]', '[STA].[PLANEDATA]', '[TMP].[AIRPORTS]',
20         '[TMP].[CARRIERS]', '[TMP].[FLIGHTDATA]', '[TMP].[LOADINGFAILURES]', '[TMP].[PLANEDATA]')
21
```

Results Messages

Search to filter items...

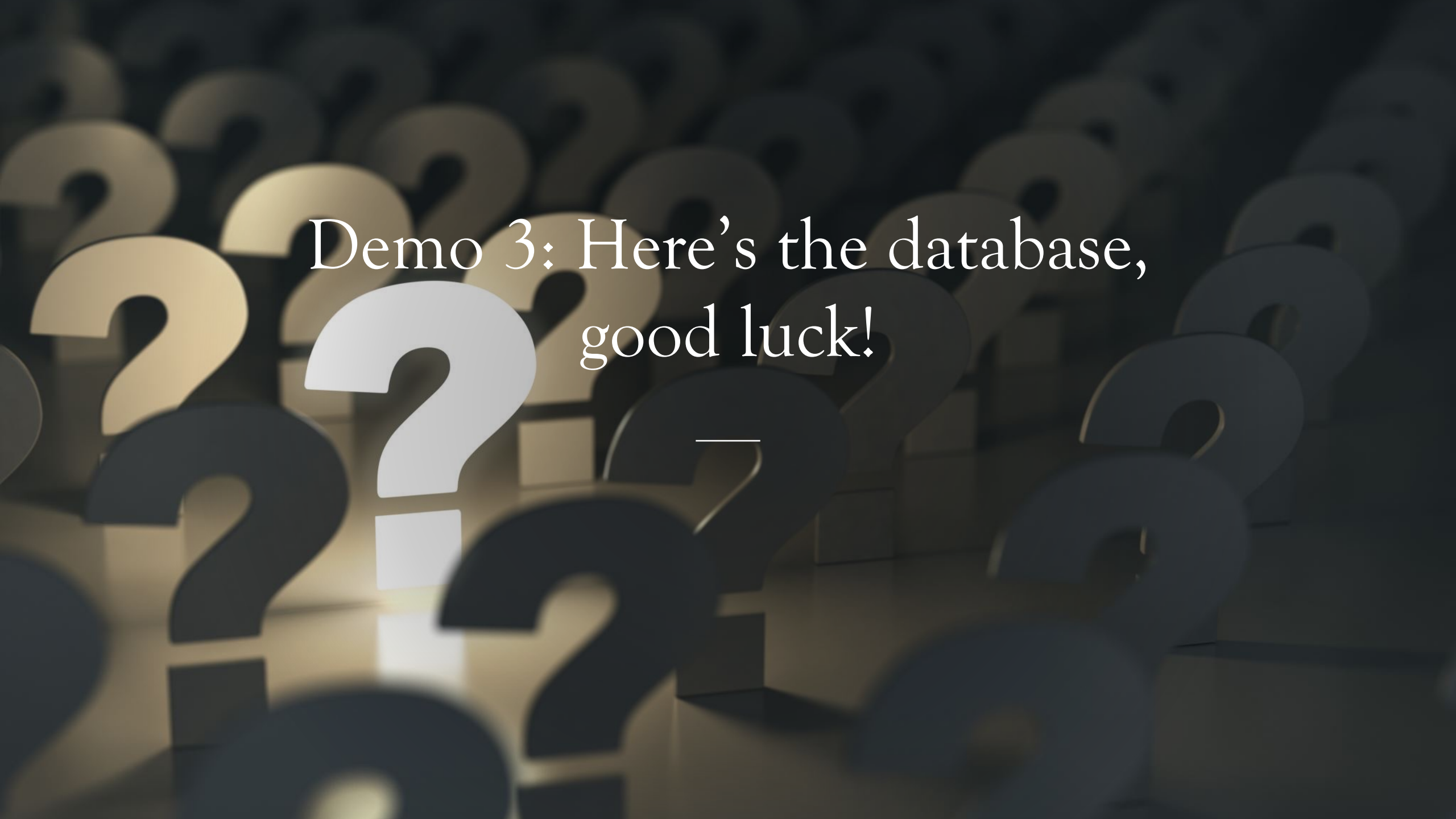
[dbo].[EmployeeHoliday]	[EmployeeID]
[dbo].[EmployeeHoliday]	[EndDate]
[dbo].[EmployeeHoliday]	[StartDate]

Finding relations between tables to answer questions

```
SELECT CONCAT(QUOTENAME(schema_name(tab1.schema_id)), '.', QUOTENAME(tab1.name)) AS [table],
       col1.name AS [column],
       CONCAT(QUOTENAME(schema_name(tab2.schema_id)), '.', QUOTENAME(tab2.name)) AS [referenced_table],
       col2.name AS [referenced_column]
FROM sys.foreign_key_columns fkc
     INNER JOIN sys.foreign_keys fk
     ON fkc.constraint_object_id = fk.object_id
     INNER JOIN sys.tables tab1
     ON tab1.object_id = fkc.parent_object_id
     INNER JOIN sys.columns col1
     ON col1.column_id = fkc.parent_column_id AND col1.object_id = tab1.object_id
     INNER JOIN sys.tables tab2
     ON tab2.object_id = fkc.referenced_object_id
     INNER JOIN sys.columns col2
     ON col2.column_id = fkc.referenced_column_id AND col2.object_id = tab2.object_id
WHERE fk.is_disabled = 0
     AND CONCAT(QUOTENAME(schema_name (tab1.schema_id))
               , '.'
               , QUOTENAME(tab1.name)) IN ('[dbo].[_TransactionIndex_31adb587-a9ae-4f15-839b-1bb9ac305049]', '[dbo].[EmployeeHoliday]',
               '[dbo].[EmployeeSalary]', '[dbo].[pivottable]', '[STA].[AIRPORTS]', '[STA].[CARRIERS]', '[STA].[FLIGHTDATA]', '[STA].[PLANEDATA]',
               '[TMP].[AIRPORTS]', '[TMP].[CARRIERS]', '[TMP].[FLIGHTDATA]', '[TMP].[LOADINGFAILURES]', '[TMP].[PLANEDATA]')
```

The background of the slide is a dark, textured surface covered with numerous question marks. Some question marks are in a light beige color, while others are in a dark charcoal color. They are scattered across the frame, creating a sense of depth and mystery. The text is centered over this background.

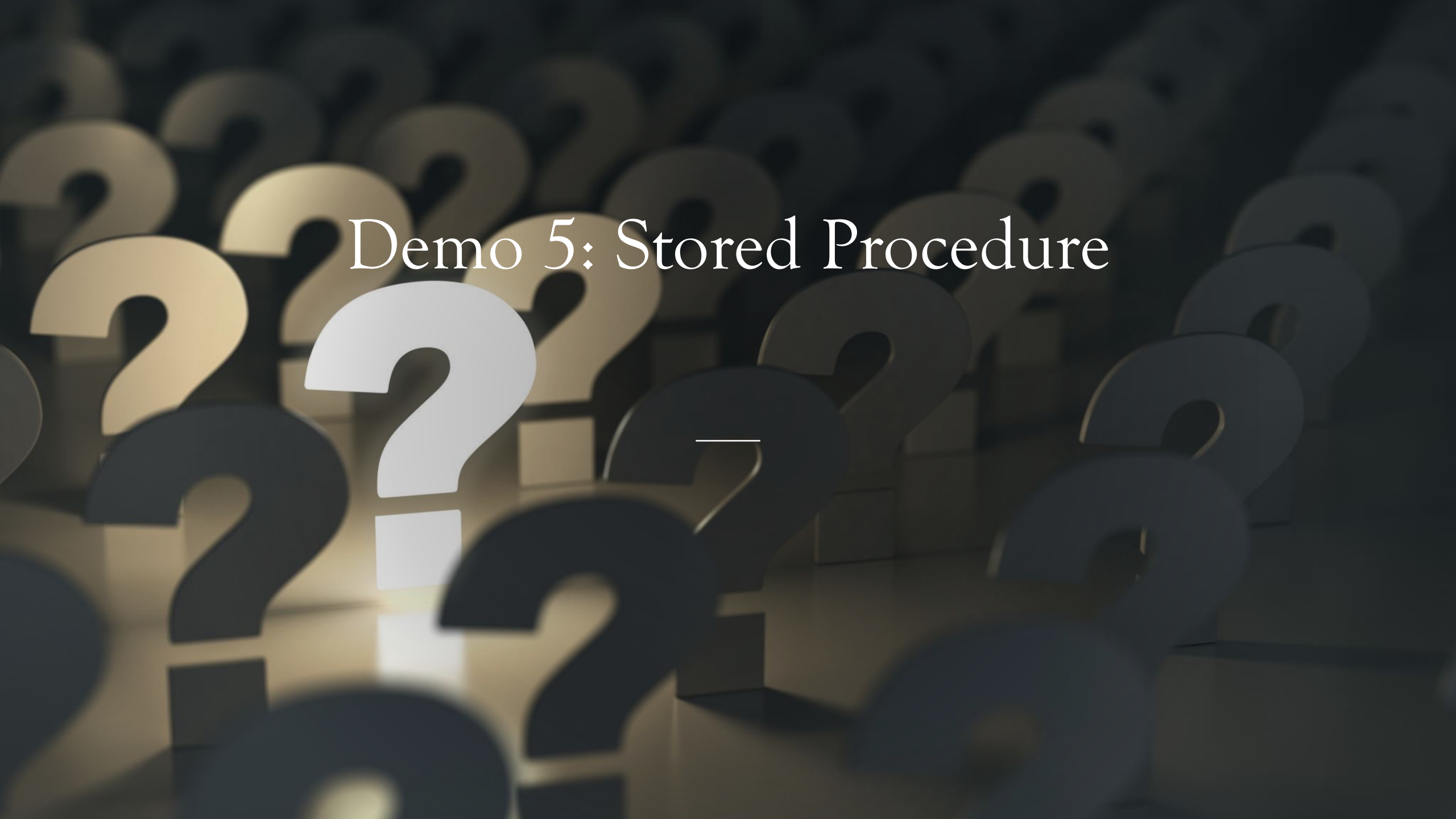
Demo 2: Starting Copilot and the different settings



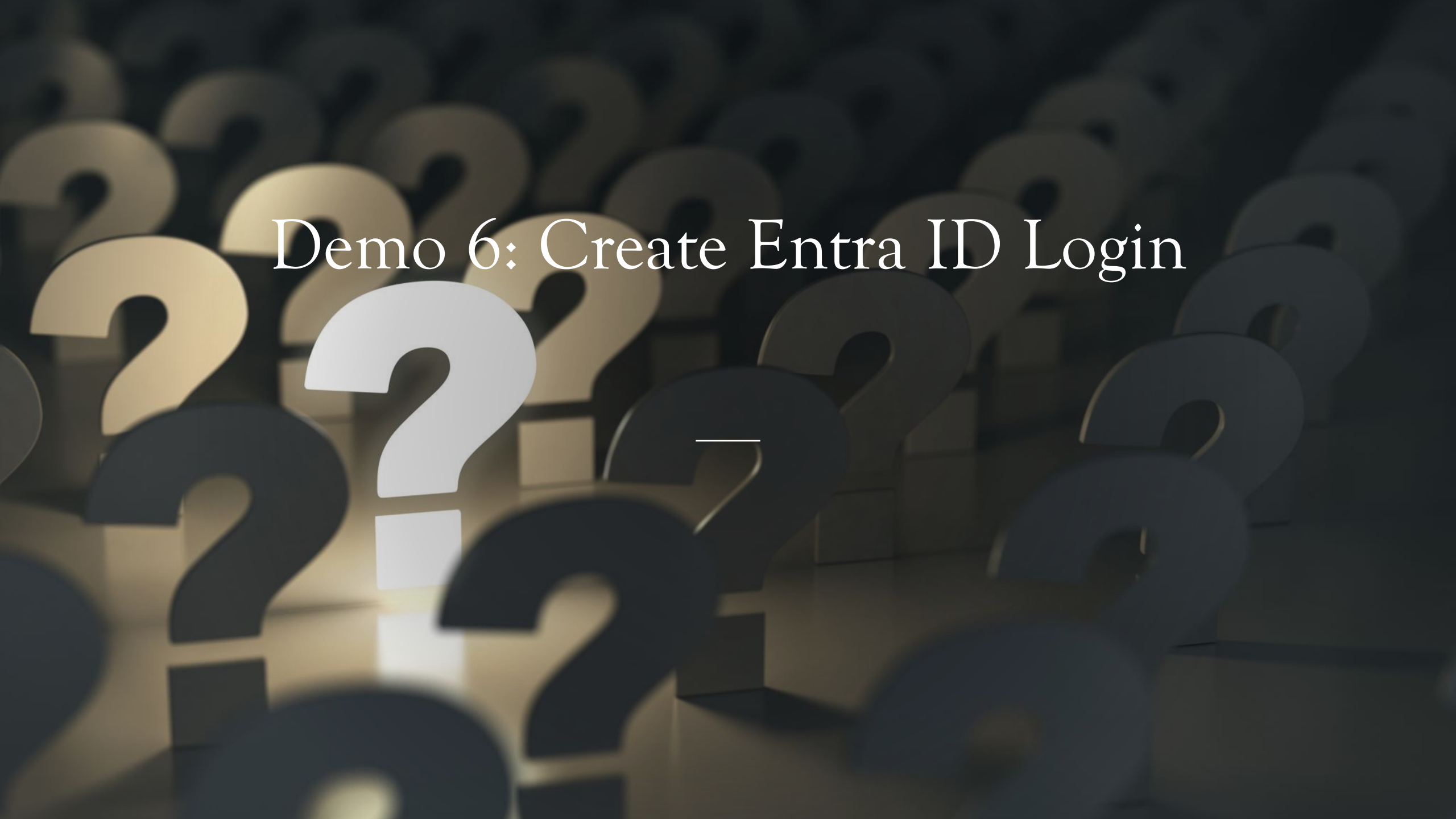
Demo 3: Here's the database,
good luck!



Demo 4: Security!

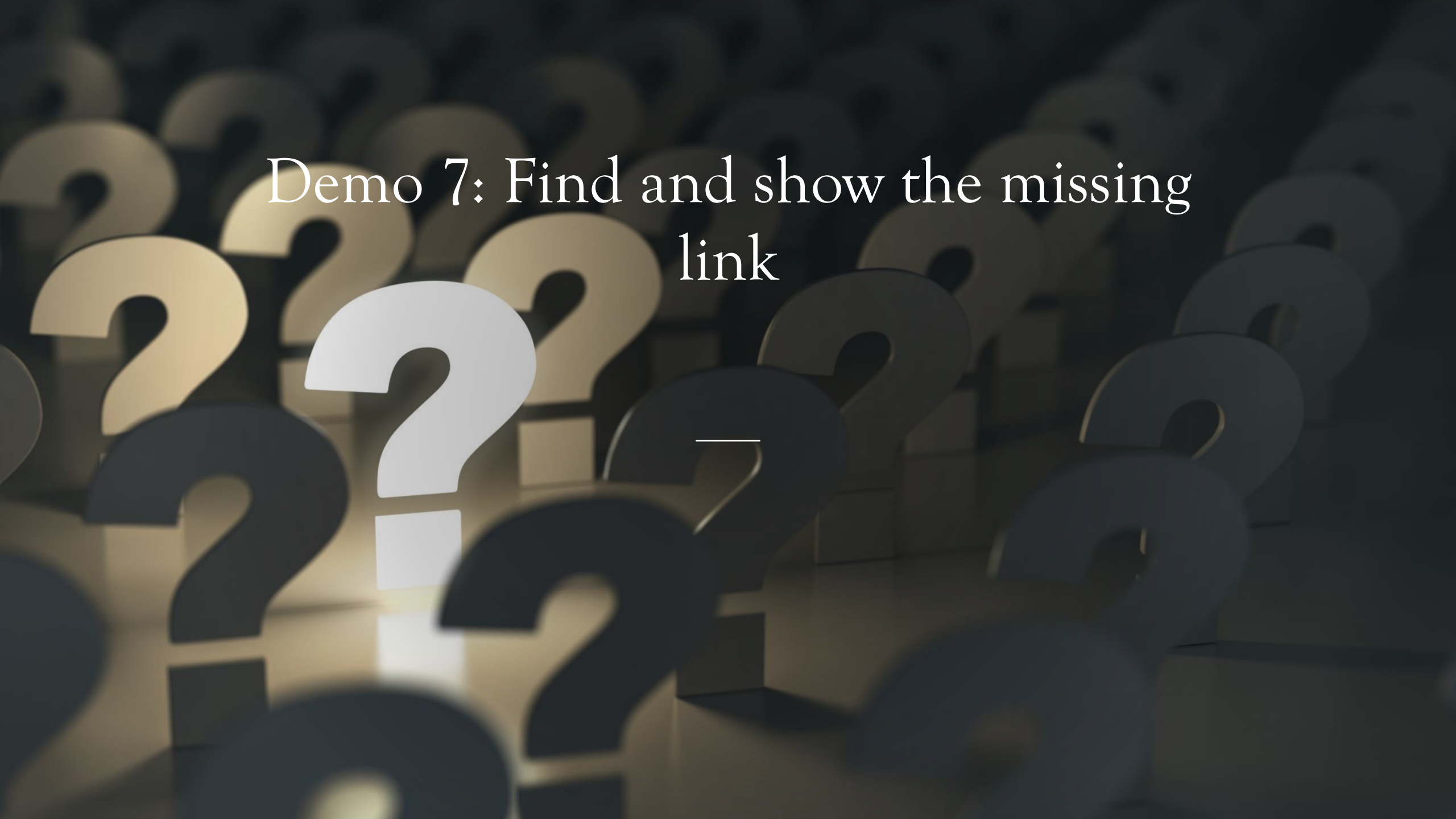
The background of the slide is a dark, textured surface covered with numerous question marks. Some question marks are in a light, metallic gold color, while others are in a dark, charcoal grey color. They are scattered across the frame, creating a sense of depth and mystery. The lighting is soft, highlighting the metallic sheen of the gold question marks.

Demo 5: Stored Procedure

The background of the slide is a dark, textured surface covered with numerous question marks. Some question marks are in a light, metallic gold color, while others are in a dark, charcoal grey color. They are scattered across the frame, creating a sense of depth and mystery. The lighting is soft, highlighting the metallic sheen of the gold question marks.

Demo 6: Create Entra ID Login

Demo 7: Find and show the missing link



Demo 8: Analyse my database,
please



Demo 9: PIVOT! PIVOT! PIVOT!



Editor Questions

How

Can

Show

Are

What

When